DIY FPV Nano Drone Project Plan

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Project Overview

This project is based on the Nano Long Range FPV drone design:

`https://makerworld.com/en/models/1181480-improved-nano-long-range-drone?from=search#profileId-1191765`

The drone is a lightweight FPV quadcopter designed to run on a single 18650 battery and be fully 3D printed.

Budget Estimate

Category	Cost (€)
Drone electronics	139
FPV Goggles	154
Controller	100
Camera	20
Filament	20
Total	**433**

Shopping List

Electronics

- Flight Controller (AIO ELRS)
- VTX (Video Transmitter)
- FPV Camera
- 4x Brushless Motors (KV $\sim$ 11500)
- LC Filter
- 5V Step-Up Converter

Power System

- 18650 Battery Holder
- 18650 Battery (Sony VTC6 recommended)

Hardware

- M2 Screws (8mm, 12mm)
- M2 Nuts
- Wires and connectors

FPV Equipment

- Radio Controller (e.g. Jumper T-Lite)
- FPV Goggles (e.g. Eachine EV100)

3D Printing

- PLA+ Filament (recommended)

3D Printed Frame



Figure 1: 3D printed drone frame

Build Process - Assembly



Figure 2: Drone assembly process

- Mount motors to frame
- Install flight controller
- Secure battery holder
- Install FPV camera and VTX

Wiring Diagram

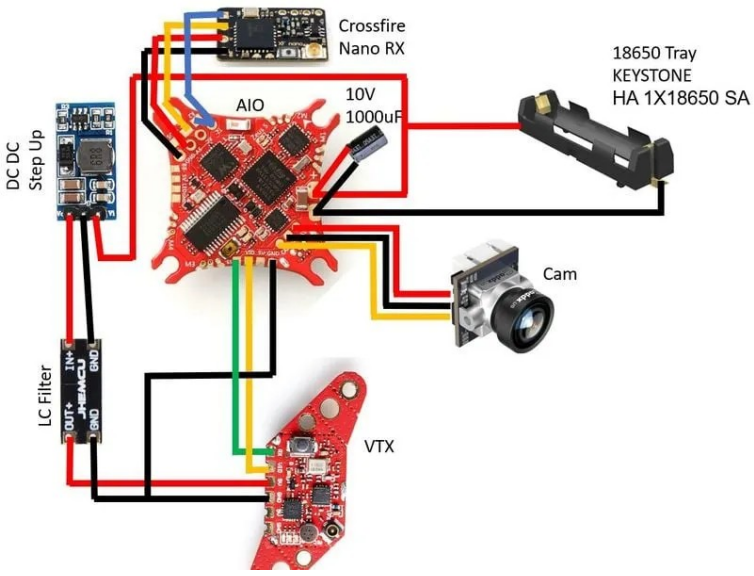


Figure 3: Complete wiring diagram of the FPV drone system

- Solder motors to flight controller
- Connect battery leads to power input
- Install LC filter and step-up converter
- Double-check polarity before powering

Final Drone Build

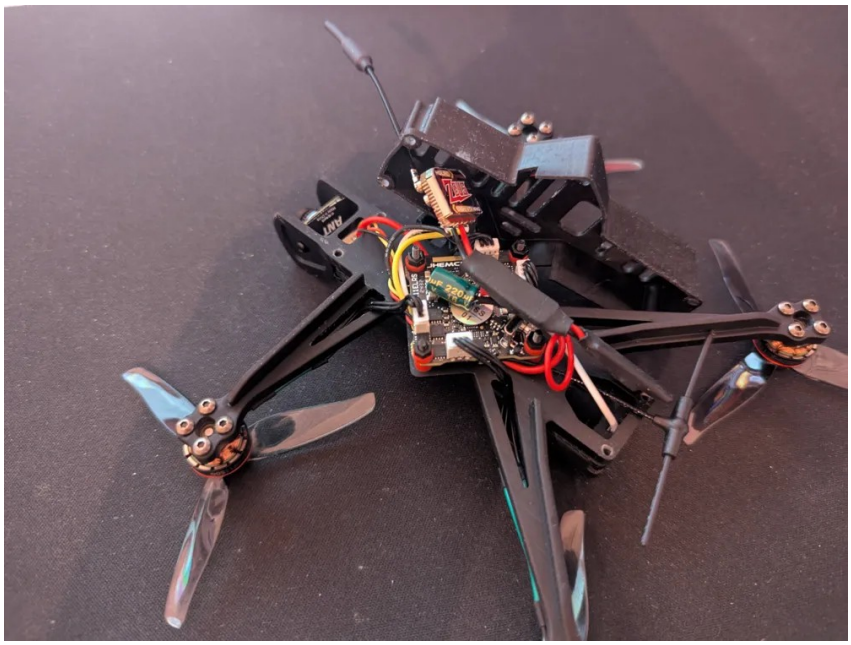


Figure 4: Final assembled FPV drone

- Perform indoor hover test
- Check stability and vibration
- Adjust PID tuning if needed
- First outdoor flight test

Important Notes

- This is an experimental DIY FPV build.
- 18650 batteries provide long flight time but limited power.
- Strong soldering skills are required.
- PLA+ is recommended for durability in printed parts.

Team Roles

- Person 1: 3D printing + mechanical assembly
- Person 2: electronics + wiring + configuration
- Both: testing and flight tuning

Future Improvements

- Upgrade to LiPo battery for better performance
- Carbon fiber frame for strength
- Improved FPV (digital system upgrade)